

Referee package (DRAFT) – STEP-5B rectangle-constant closure: the theorem-grade rectangle prefactor

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

This referee document concerns the STEP-5B rectangle-constant closure: the theorem-grade prefactor $K(n)$ controlling the layer budget. **Scope:** the rectangle constant and the official threshold. **Not claimed:** the alternative 20/9 incidence route is provisional (constant unpinned) and is not relied upon.

2. The result

(i) THEOREM (operator verdicts #9/#10): $K(n) \leq 8 + 4\sqrt{14}\kappa^4\sqrt{n}$, with 1.59×10^5 the OFFICIAL threshold. (ii) The 20/9 incidence route (2.2×10^{10}) is provisional with an unpinned constant and is recorded but not load-bearing.

3. Structure of the argument

The rectangle prefactor is bounded by the closed-form $8 + 4\sqrt{14}\kappa^4\sqrt{n}$ (the operator-verified rectangle theorem), giving the official threshold 1.59×10^5 that the layer budget must clear. The incidence-geometry alternative is documented for cross-check but its constant is not pinned, so it is not used in the closure.

4. Numerical reproduction

The beyond-layer Gershgorin bound script reproduces the rectangle constant and the 1.59×10^5 threshold with asserts; run it from the bundle.

5. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) *Attack the scope.* The result is confined to the scope stated in Sec. 1; a referee may push that boundary, which is fair — the package makes no claim outside it, and the precise scope is in the footer.
- (β) *Attack the tier / over-claim.* No prose here is stronger than the registered grade (footer no-overclaim line); the status is REVIEW DRAFT, NOT operator-confirmed, and the claim-ledger tier is the arbiter.
- (γ) *Attack reproduction / self-containment.* Each reproduction script carries self-test asserts and ships in the folder’s bundle with its transitive dependencies (imports and runtime file reads, resolver v1.3.0); a referee runs them and diffs against `expected/`.

Quantitative sanity check (mandatory): the falsifier (footer) is concrete and pre-registered, so the claim is refutable by a single explicit construction; and the numerical values it cites are reproduced by the folder's scripts (all asserts pass).

6. Result footer

Result ID:	B5-BEYOND-LAYER-BOUND / STEP-5B
Precise statement:	rectangle-constant referee package (DRAFT) THEOREM (operator verdicts #9/#10): $K(n) \leq 8 + 4 \sqrt{14} \kappa^4 \sqrt{n}$; 1.59e5 is the OFFICIAL threshold for the layer budget. The 20/9 incidence route (2.2e10) is provisional (constant unpinned) and not load-bearing.
Scope:	the STEP-5B rectangle prefactor $K(n)$ and official threshold. Theorem-grade rectangle constant.
Evidence grade:	theorem-grade rectangle constant (T5/T6); EXECUTED.
Reproduction command:	python codes/vacuum/beyond_layer_gershgorin_bound.py
Falsifier:	a configuration where $K(n)$ exceeds $8 + 4 \sqrt{14} \kappa^4 \sqrt{n}$; or the layer budget failing the 1.59e5 threshold.
Tier before / after:	DRAFT referee package; rectangle constant T5/T6. NOT operator-confirmed.
No-overclaim statement:	the rectangle constant is theorem-grade; the 20/9 incidence route is provisional and not relied upon.
Next required action:	operator review + confirm -> PUBLISHED bundle.